THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** biodiversity	diversity of life
2	**(c)** habitat — land, water and air	4th century BCE
3	**(a)** autotrophic prokaryotes	photosynthetic
4	**(a)** absorbing nutrients from dead organic matter	saprotrophic
5	**(d)** algae (thallophyta)	found in water
6	**(c)** gymnosperms	naked seeds
7	**(a)** non-chordates	invertebrates
8	**(c)** Annelida	segmented body
9	**(b)** fishes	aquatic
10	**(a)** Carolus Linnaeus	binomial nomenclature

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Variety of life forms on the Earth (1); it supports food webs and ecosystem stability (1).	1+1
12	Grouping of organisms on the basis of similarities and differences (1); to study the vast diversity systematically (1).	1+1
13	Body structure, cell structure, mode of nutrition, habitat, reproduction, genetic similarity (any four, ½ each).	2
14	Monera, Protista, Fungi, Plantae and Animalia (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Rich diversity supports food webs and nutrient cycling (1.5); mangrove diversity reduced destruction in the 1999 Odisha cyclone (1.5).	1½+1½
16	Regions rich in the number of species and under threat (1.5); Western Ghats and Indo-Burma (1.5).	1½+1½
17	Makes study easier; shows relationships; helps identify and name new organisms; supports conservation (1+1+1).	1+1+1
	OR: Aristotle: habitat; two kingdoms: plants and animals; later more kingdoms; Whittaker's five (3).	3
18	Cell type (prokaryotic/eukaryotic), body organisation, mode of nutrition, cell wall, reproduction (3).	3
	OR: Prokaryotic organisms such as bacteria and cyanobacteria (1.5); Lactobacillus and Rhizobium are useful; bacteria produce biogas (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Variety of life (1). (ii) Maintains ecosystems, food, medicines and stability (2). (iii) Decomposers recycle nutrients; plants provide food and oxygen (2).	1+2+2
	OR: (i) A small area supports great diversity (1). (ii) They use different nesting cavities and fruits (1). (iii) They have many threatened species (1). (iv) Protect forests, control hunting (2).	1+1+1+2
20	(i) To study diversity (1). (ii) Cell type, body organisation, nutrition, habitat, reproduction (2). (iii) Five kingdoms with one feature each (2).	1+2+2
	OR: (i) Euglena and Paramecium: Protista; Spirogyra: Plantae (thallophyta) (1). (ii) Euglena and Spirogyra (1). (iii) Paramecium (1). (iv) Spirogyra (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Western Ghats	rich in species
2	**(d)** R. H. Whittaker	1969
3	**(d)** Protista	unicellular eukaryotes
4	**(a)** an alga and a fungus	bioindicators of air pollution
5	**(c)** bryophytes	amphibians of the plant kingdom
6	**(a)** angiosperms	flowering plants
7	**(b)** Porifera	pores
8	**(a)** Arthropoda	insects, spiders, crabs
9	**(b)** birds	warm-blooded
10	**(c)** with a capital first letter, in italics	e.g. Homo sapiens

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They photosynthesise and released oxygen, making the Earth suitable for life (2).	2
12	Unicellular eukaryotes; with or without cell wall; e.g. amoeba (1+1).	1+1
13	They feed on dead organic matter and recycle nutrients into the soil (2).	2
14	A symbiotic association of an alga and a fungus (1); bioindicator of air pollution (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Useful: Lactobacillus in curd, Rhizobium nitrogen fixation, decomposition, biogas (1.5); harmful: cause diseases (1.5).	1½+1½
16	Mostly multicellular eukaryotes with chitin walls (1); absorb food (1); yeast, mushrooms, Aspergillus (1).	1+1+1
17	Alga and fungus live together; the alga photosynthesises and the fungus gives protection (2); they change colour with pollutants (1).	2+1
	OR: Algae: thallus, aquatic; bryophytes: rhizoids, no vascular tissue; pteridophytes: vascular tissue, true roots, stems, leaves (1+1+1).	1+1+1
18	They live on land but need water for reproduction (3).	3
	OR: Seeds protect the embryo and aid dispersal (1.5); fruits enclose seeds and help dispersal by animals and wind (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Prokaryotes vs unicellular eukaryotes (2). (ii) Nucleus absent/present, size, examples (2). (iii) Lactobacillus and Rhizobium (1).	2+2+1
	OR: (i) Chitin walls, mostly multicellular, no chlorophyll (2). (ii) Absorb from dead organic matter or hosts (1). (iii) Alga + fungus; bioindicators (2).	2+1+2
20	(i) Lichens (1). (ii) Alga and fungus (1). (iii) Alga: food; fungus: protection (1). (iv) They change colour or disappear (2).	1+1+1+2
	OR: (i) Thallophyta (Spirogyra), Bryophyta (moss), Pteridophyta (fern), Gymnosperms (pine), Angiosperms (mango) (3). (ii) Features of each (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.